LOCALISATIONS OF A NOETHERIAN ZARISKI PAIR AT ELEMENTS OF THE IDEAL ARE JACOBSON RINGS

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Allowed standard prerequisites (editorial).

- The criterion for a Noetherian ring to be Jacobson via dimension one primes and infinite spectra. Algebra, Lemma \ref{algebra-lemma-noetherian-dim-1-Jacobson}, cited at proof lines 2353-2354. Is the criterion that the proof verifies.
- The Zariski pair property: the ideal is contained in every maximal ideal. the hypothesis that the pair is a Zariski pair, used at proof lines 2363-2364. Makes the maximal ideal containing the chosen prime distinct from it.
- The spectrum of the localisation of a two-dimensional Noetherian local domain at an element is infinite. Algebra, Lemma \ref{algebra-lemma-Noetherian-local-domain-dim-2-infinite-opens}, cited at proof lines 2367-2368. Delivers the infinite spectrum required by the criterion.

Source reference. Original lemma 0GED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20473–20473. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00KX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 767–767. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02IG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3299–3299. Navigation pointer to a statement the

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Lemma 0.2. *Let (A, I) be a Zariski pair with A Noetherian. Let $f \in I$. Then A_f is a Jacobson ring.*

Proof. We will use the criterion of Algebra, Lemma 00KX. Let $\mathfrak{p} \subset A$ be a prime ideal such that $\mathfrak{p}_f = \mathfrak{p}A_f$ is prime and not maximal. We have to show that $A_f/\mathfrak{p}_f = (A/\mathfrak{p})_f$ has infinitely many prime ideals. After replacing A by $A/\mathfrak{p}$ we may assume A is a domain, $\dim A_f > 0$, and our goal is to show that $\mathrm{Spec}(A_f)$ is infinite. Since $\dim A_f > 0$ we can find a nonzero prime ideal $\mathfrak{q} \subset A$ not containing f. Choose a maximal ideal $\mathfrak{m} \subset A$ containing $\mathfrak{q}$. Since (A, I) is a Zariski pair, we see $I \subset \mathfrak{m}$. Hence $\mathfrak{m} \neq \mathfrak{q}$ and $\dim(A_\mathfrak{m}) > 1$. Hence $\mathrm{Spec}((A_\mathfrak{m})_f) \subset \mathrm{Spec}(A_f)$ is infinite by Algebra, Lemma 02IG and we win. □